

Customer	
INT	Customer_ID
VARCHAR100	Name
VARCHAR150	Email
VARCHAR15	Phone
VARCHAR255	Address

places

Sale	
INT	Sale_ID
INT	Customer_ID
DATETIME	Sale_Date
DECIMAL10_2	Total_Amount

generates

Bill	
INT	Bill_ID
INT	Sale_ID
DECIMAL10_2	Subtotal
DECIMAL10_2	Tax
DECIMAL10_2	Discount
DECIMAL10_2	Grand_Total
VARCHAR50	Payment_Method

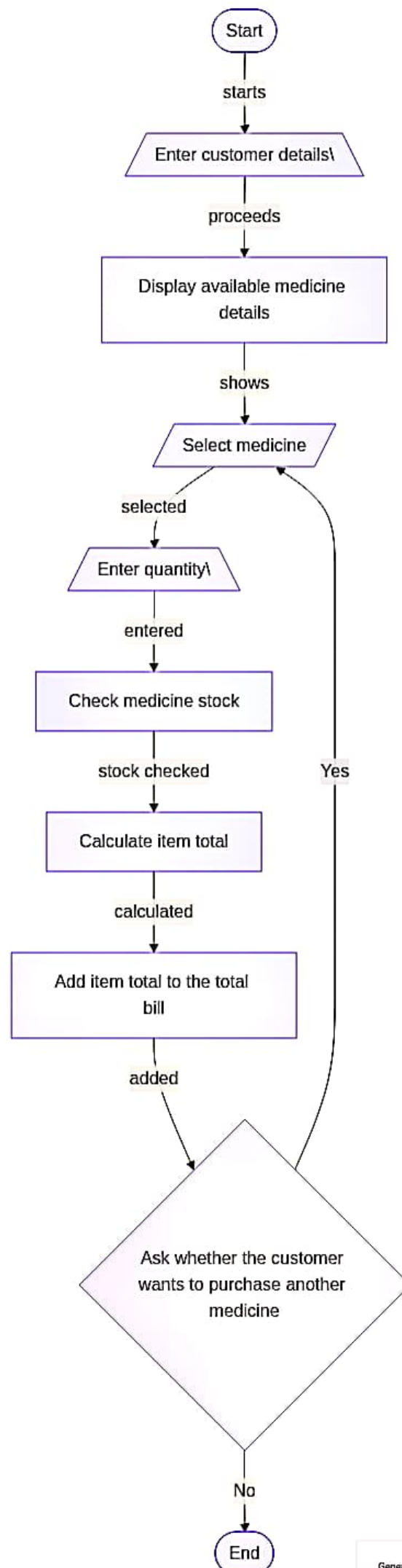
Medicine

INT	Medicine_ID
VARCHAR100	Medicine_Name
VARCHAR50	Category
DECIMAL10_2	Price
INT	Stock_Quantity

included_in

Sale_Item	
INT	Sale_ITEM_ID
INT	Sale_ID
INT	Medicine_ID
INT	Quantity
DECIMAL10_2	Unit_Price
DECIMAL10_2	Line_Total

contains



Algorithm for Medicine Management System

Input:

- Customer ID
- Customer Name
- Contact Number
- Medicine ID
- Medicine Name
- Medicine Price
- Quantity

Processing:

Step 1: Accept customer details.

Step 2: Display available medicine details.

Step 3: Select a medicine.

Step 4: Enter the required quantity.

Step 5: Check whether sufficient medicine stock is available.

Step 6: If stock is insufficient, display an appropriate message and select another medicine.

Step 7: Calculate the item total using:

$$\text{Item Total} = \text{Medicine Price} \times \text{Quantity}$$

Step 8: Add the item total to the total bill using:

$$\text{Total Bill} = \text{Total Bill} + \text{Item Total}$$

Step 9: Ask whether the customer wants to purchase another medicine.

Step 10: If Yes, repeat the medicine selection and quantity process.

Step 11: If No, generate the final bill.

Step 12: Update the medicine stock after the sale.

Step 13: Display customer details, selected medicines, quantities, prices and final bill amount.

Output:

- Customer details
- Medicine details
- Quantity purchased
- Individual item totals
- Final total bill
- Sale completed message